

How a photo of a dog to a three-year-old child, and she or he will be able to tell you it is a dog, and that may not be considered remarkable. But if a machine can tell you the same thing, it would be considered remarkable. This is because for a machine to achieve that feat would require a form of artificial intelligence (AI) development called deep learning. The machine has to be trained using thousands of dog images, otherwise it will merely perceive it as a blob and give an incorrect answer. In the above example, unlike the machine, the child can learn things intuitively and unsupervised, and it comes as no surprise when she or he can easily recognise the animal.

Not Hotdog app seems farcical and absurd, but it has a very interesting factual and practical aspect to it. This app is based on cutting-edge AI and computer vision technologies that can identify a hotdog from other food items or objects. It was developed for fun and experimentation. But with advancements in modern technology, we can use AI algorithms to easily identify animals, objects, places and people for use in critical applications.

It is easy to build an AI app like Not Hotdog for fun and experimentation, but what does it take to build a mission-critical AI application that you can trust to help run a business or accomplish a task successfully? Let us take a look at a few cases of AI in mission-critical applications in this article.

If Not Hotdog app does not give an answer, it will not matter much as it is not critical. But think of a situation where a computer vision system outputs wrong information in skin cancer detection, or a self-driving car is unable to detect an object in front of it, or a situation where the visual object detection system fails to detect the enemy in a military operation?

The results would be catastrophic and cause irreparable damage.



Figure 1: Screenshot of Not Hotdog app (Credit: www.oreilly.com)

These are critical applications where accuracy, timing and quality are extremely important.

AI technology related to computer vision, image processing, machine learning and deep learning has many applications across industries.

In consumer electronics, these applications provide face recognition and autofocus to camera-enabled devices.

In the public sector, these can be used to identify dangerous driving situations with traffic monitoring cameras.

In healthcare, these can be used for medical imaging and diagnosis.

In retail, these can be used to spot malicious activities in stores.

In agriculture, these can be used to determine the health of crops in the field.

All of these applications are critical to core businesses and processes.

In critical applications, quality and quantity of input data are important to obtain correct results. Deep learning based on a deep neural network, a class of machine learning algorithms, is important in mission-critical applications. Deep learning uses multiple layers to progressively extract higher-level features from raw data input, enhancing accuracy and giving more reliable results.

Key differences in mission-critical AI

The context and levels of criticality may differ from industry to industry, but there are a few parameters common to all such

applications involving AI technology. Mission-critical, time-critical, life-critical and safety-critical applications require fast identification, timely action, data delivery, reliable notification and warning messages. Key differences in mission-critical AI applications are accuracy, service-level agreement (SLA), scalability and security.

Accuracy. Mission-critical AI applications need significant and better accuracy. This means that training data needs to be greater in quality and quantity. Actually, quantity of data you need depends on complexity of algorithms for a particular problem.

SLA: All core business processes have strict SLAs. An SLA is a commitment between a service provider and a client. The commitment includes quality, availability and responsibilities. Services should be provided to the customer as agreed upon in the contract. For example, when an image is fed to a model for prediction, as per requirement, the model should not take much time to predict the correct result. If there is a delay, it is not following a strict SLA.

SLAs may include many components, from a definition of services to termination of agreement.

Scalability: Besides SLAs, architecture has to be scalable. High accuracy and strict SLAs need an architecture that can effortlessly scale for both computing and storage.

Security: This is one of the prime concerns in AI technology. Certain applications present security and data privacy concerns. Data may include sensitive information that has to be stored securely with enterprise-grade security features. In mission-critical AI applications, strict security measures are taken into account and only authorised users have access to information.

AI components in mission-critical applications

Essential components for mission-critical AI applications include datasets